Second Law of Thermodynamics & Entropy

Recall the first law of thermodynamics on energy conservation, being that energy cannot be created or destroyed, but can be converted from one form to another.

However, a process that follows the first law may not necessarily be accurate to real world situations, as the first law only governs the amount of energy. Instead, the second law of thermodynamics ensures that the flow of energy is accurate.

The second law of thermodynamics states that any energy transfer or conversion proceeds in a certain direction, from a concentrated region to a diluted region. All processes must obey both laws.

Entropy

The second law of thermodynamics can also be talked about in terms of entropy. Entropy is a measure of the randomness, or disorder in a system, and a process will tend to an entropic state, or a state from lower entropy to higher entropy. Entropy can also be said as the amount of energy unavailable to do work. Hence, entropy is also called the arrow of time, as it controls the direction a process goes from two states.

Statements of the Second Law

Unlike the first law, which can be easily expressed, a concise statement for the second law is harder to find. However, there are two common ways to state the second law, both in the context of cyclic devices:

Kelvin-Planck Statement

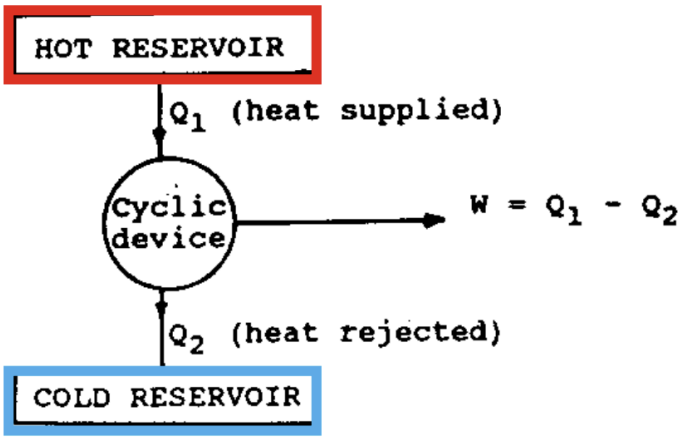
The Kelvin-Planck Statement states that it is impossible to have a cyclic device that extracts heat from a reservoir and does an equivalent amount of work on the surroundings.

Consider a cyclic device, like a heat engine or heat pump, that takes in heat to do work.

The source of the heat input is known as a hot reservoir, and what the Kelvin-Planck statement states is that if the cyclic device takes in Q kJ of

heat from the hot reservoir, it is impossible for it to output Q kJ of work on the surroundings.

Instead, some heat has to be rejected into a cold reservoir.

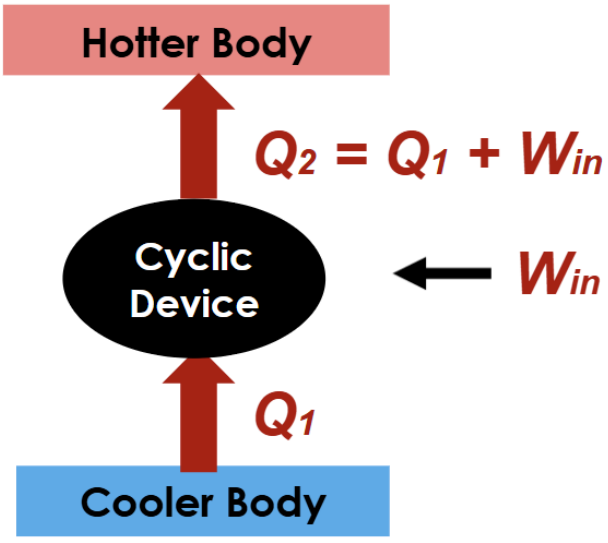


Clausius' Statement

Clausius' Statement states that it is impossible to have a cyclic device that can have no other effects other than the transfer of heat from a cooler body to a hotter body.

Again, consider a cyclic device like a heat pump or a refrigerator.

A cyclic device is unable to shift an amount of heat from the cooler body (e.g. inside of a fridge) to the hotter body (surroundings) on its own, and needs some work input for it to transfer the energy.



Reversibility

A reversible process is an ideal process that can be restored into its original state. This means that the process must not dissipate any excess energy into the surroundings, like via friction, as the reverse of the process cannot take in the dissipated energy, hence making it an impossible ideal.

As such, most if not all processes are irreversible, as they degrade over time. However, a process can be improved such that it is as close to reversible as possible, increasing the efficiency of it as less heat will be dissipated.

Some criteria for a reversible process is that it has to be frictionless, and the pressure and temperature difference between the fluid and the surroundings must be infinitely small.

Thermodynamic Definition of Entropy

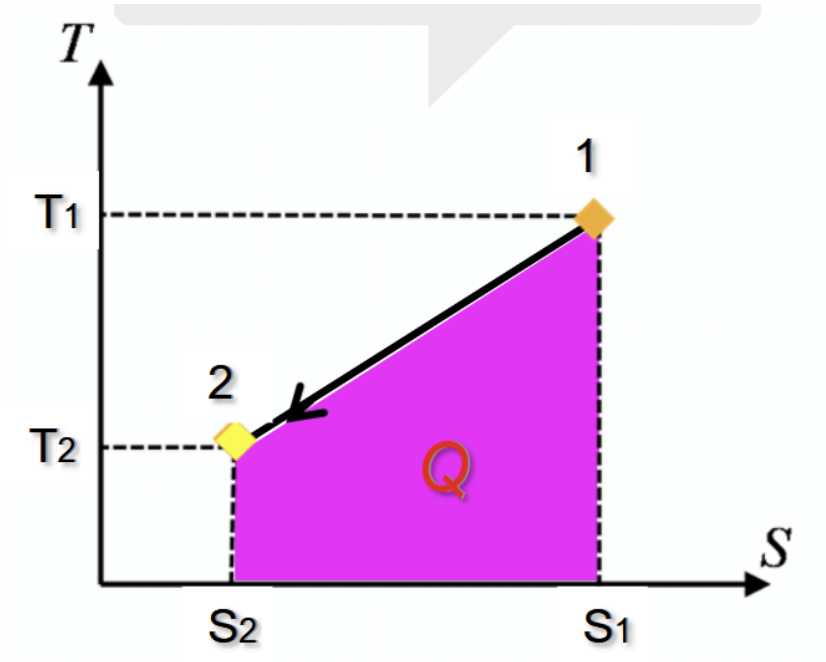
Entropy is denoted by S, in units kJ/K. Specific entropy is also denoted by s, in units kJ/kgK.

Typically, we care more about the change in entropy in a process. For a reversible process,

$$\Delta S = S_2 - S_1 = \int_{T_1}^{T_2} \frac{1}{T}\, dQ$$

Using this, we can determine the change in entropy for different processes.

Similarly, for T-S diagrams, the area under the process line is the heat transferred in the (reversible) process.



For reversible processes, entropy is transferred between the system and surroundings, so a decrease in entropy in the system causes an increase in entropy in the surroundings. Irreversible processes are different, in that a decrease in entropy in the system will cause an even greater increase in entropy in the surroundings, while a increase in entropy will cause a smaller decrease in entropy in the surroundings. Both cases cause a total entropy generation in the universe. The idea of entropy continuously increasing in the universe due to real processes is also called the principle of increase in entropy.

Change in Entropy for Ideal Gas Processes

Isochoric Processes

Recall the first law of thermodynamics, $Q - W = \Delta U$. The statement can be rewritten as

$$\begin{gathered} Q = U_2 - U_1 + W \\ dQ = d(U_2 - U_1 + W) \\ dQ = d(mC_vT_2 - mC_vT_1 + pV) \\ dQ = mC_vdT + pdV \end{gathered}$$

As isochoric processes do not do any work, $dQ = mC_vdT$.

Hence,

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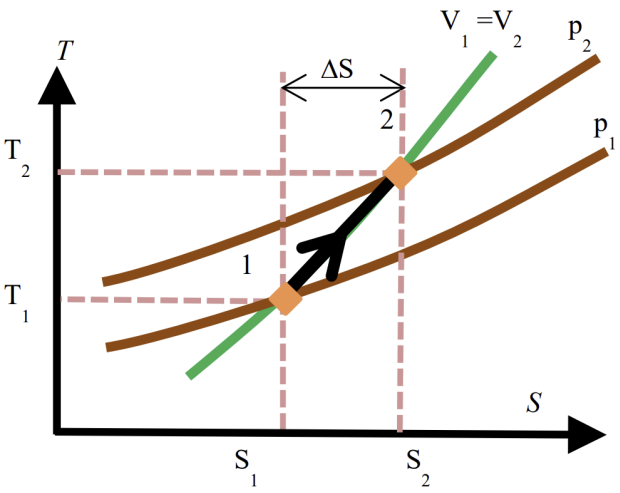
$$
\begin{aligned}
\Delta S &= \int_{T_1}^{T_2} \frac{1}{T}\, dQ \\
&= \int_{T_1}^{T_2} \frac{1}{T}\, mC_v dT \\
&= mC_v \int_{T_1}^{T_2} \frac{1}{T}\, dT \\
&= mC_v \left[\ln(T)\right]_{T_1}^{T_2} \\
&= mC_v \ln\left(\frac{T_2}{T_1}\right)
\end{aligned}
$$

Using the equation of state:

$$
\frac{P_1}{T_1} = \frac{P_2}{T_2} \Rightarrow \frac{T_2}{T_1} = \frac{P_2}{P_1}
$$

$$
\therefore \Delta S = mC_v \ln\left(\frac{T_2}{T_1}\right) = mC_v \ln\left(\frac{P_2}{P_1}\right)
$$

On a TS diagram, the process is represented by a curved line that stays along a constant volume line and is bounded by two constant pressure lines.



Shown is an increase in pressure, as the temperature is also increasing.

Notice how the intersections (which are also states 1 and 2) between the constant lines represent the state at pressure P1 and volume V1, and pressure P2 and volume V1.

**Drawings of the constant lines are needed i think!!!!!!!!!!!!

Isobaric Processes

Recall that the heat transferred for isobaric processes is $Q = H_2 - H_1$. Hence,

$$dQ = dH_2 - dH_1$$
$$dQ = d(mC_pT)$$
$$dQ = mC_pdT$$

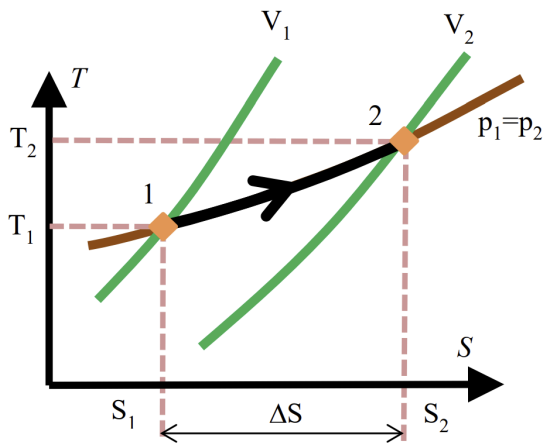
Using this, we can find the change in entropy:

$$\Delta S = \int_{T_1}^{T_2} \frac{1}{T}\, dQ$$
$$= mC_p \int_{T_1}^{T_2} \frac{1}{T}\, dT$$
$$= mC_p \left[\ln(T)\right]_{T_1}^{T_2}$$
$$= mC_p \ln\left(\frac{T_2}{T_1}\right)$$

Using the equation of state:

$$\frac{V_1}{T_1} = \frac{V_2}{T_2} \Rightarrow \frac{T_2}{T_1} = \frac{V_2}{V_1}$$
$$\therefore \Delta S = mC_p \ln\left(\frac{T_2}{T_1}\right) = mC_p \ln\left(\frac{V_2}{V_1}\right)$$

On the TS diagram, the process stays on the constant pressure line and is bounded by the constant volume lines. It is also a gentler slope than the isochoric process.



Isothermal Processes

Isothermal processes do not have any change in internal energy, hence

$$
\begin{aligned}
Q - W &= 0 \\
Q &= W \\
dQ &= dW \\
dQ &= pdV
\end{aligned}
$$

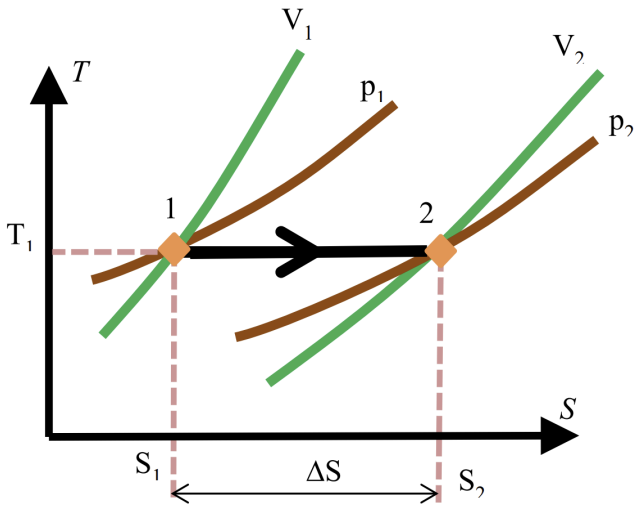
Hence,

$$
\begin{aligned}
\Delta S &= \int_{T_1}^{T_2} \frac{1}{T}\, dQ \\
&= \int_{T_1}^{T_2} \frac{p}{T}\, dV \\
&= \int_{T_1}^{T_2} \frac{mRT}{VT}\, dV \\
&= \int_{T_1}^{T_2} \frac{mR}{V}\, dV \\
&= mR[\ln(V)]_{T_1}^{T_2} \\
&= mR \ln\left(\frac{V(T_2)}{V(T_1)}\right) \\
&= mR \ln\left(\frac{V_2}{V_1}\right)
\end{aligned}
$$

Using the equation of state:

$$
\begin{aligned}
P_1V_1 = P_2V_2 \Rightarrow \frac{V_2}{V_1} &= \frac{P_1}{P_2} \\
\therefore \Delta S = mR \ln\left(\frac{V_2}{V_1}\right) &= mR \ln\left(\frac{P_1}{P_2}\right)
\end{aligned}
$$

On a TS diagram, the process is a horizontal line, as temperature does not change.



Polytrophic Processes

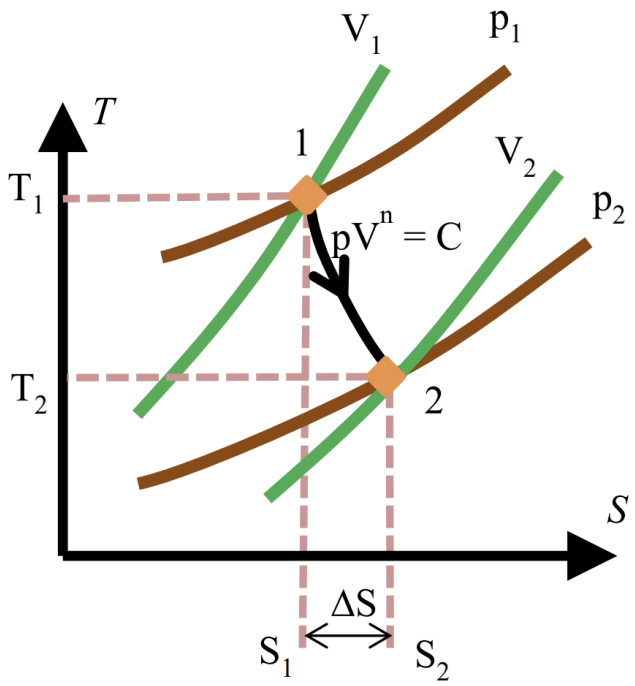
Recall the equation:

$$dQ = mC_v dT + pdV$$

Applying into the change in entropy:

$$\begin{aligned}
\Delta S &= \int_{T_1}^{T_2} \frac{1}{T}\, dQ \\
&= \int_{T_1}^{T_2} \frac{1}{T}\, mC_v dT + pdV \\
&= \int_{T_1}^{T_2} \frac{1}{T}\, mC_v dT + \int_{T_1}^{T_2} pdV \\
&= mC_v \int_{T_1}^{T_2} \frac{1}{T}\, dT + mR \int_{T_1}^{T_2} \frac{1}{V}\, dV \\
&= mC_v \ln\left(\frac{T_2}{T_1}\right) + mR \ln\left(\frac{V_2}{V_1}\right)
\end{aligned}$$

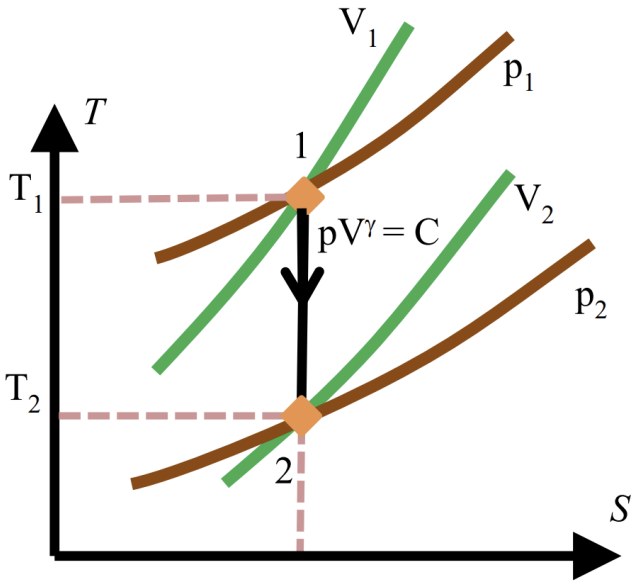
On the TS diagram, the process is represented by a curve



Reversible Adiabatic Processes

Recall that an adiabatic process has $Q = 0$, hence $dQ = 0$. Thus $\Delta S = 0$.

As such, the TS diagram of this process is a vertical line, as entropy does not change.



$$S_1 = S_2$$

$$\Delta S = 0$$

A process that keeps ΔS at 0 is also called an isentropic process, or a process of constant entropy. A reversible adiabatic process is an isentropic process, but an irreversible adiabatic process is not one.

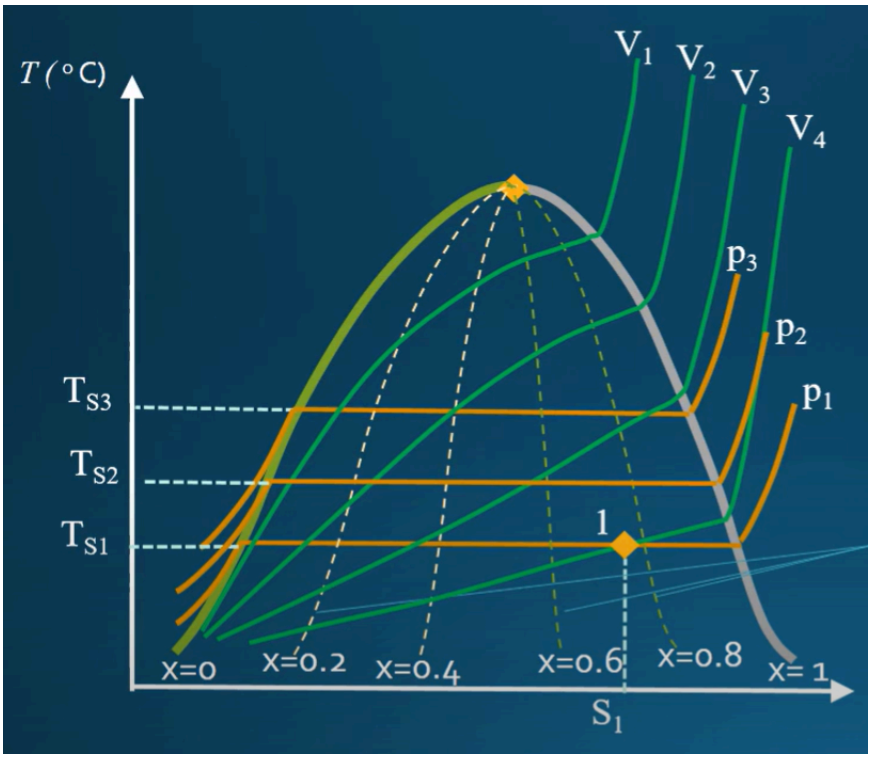
Change in Entropy for Steam Processes

For steam processes, ΔS can be found from the steam tables, and also by using the formula

$$s = s_f + x s_{fg}$$

for the different states, and finding the difference, as $\Delta S = S_2 - S_1$

Shown is the TS diagram of the heating of water from subcooled water to superheated steam.



Notice the main boundary determines when it transitions between the saturated states.

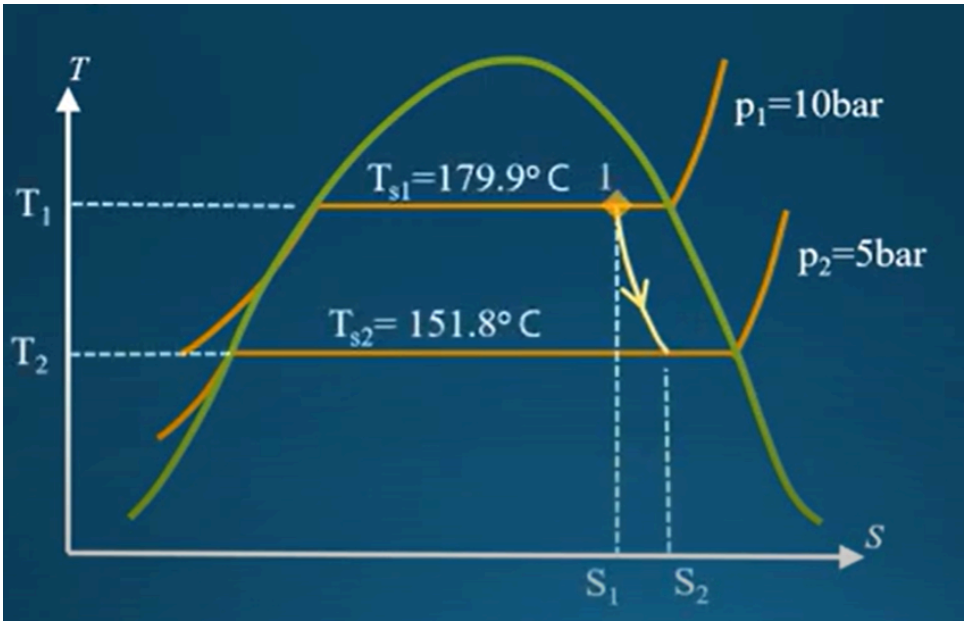
Notice the constant pressure and volume lines, which can be used to locate and draw different states and steam processes.

Like isochoric and isobaric ideal gas processes, they follow the respective constant line and are bounded by the other lines. Note that processes that

cross the boundary curve will still follow the shape of the constant line, even when bent.

Isothermal processes again are horizontal lines, as temperature does not change, however is bounded by the respective pressure lines.

Polytropic processes are drawn as bounded between two pressure or volume lines, depending on the setup, but are still drawn as curves:



while isentropic processes are drawn as vertical lines, bounded similarly.